

| 1. Algebra and functions |                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1.1                      | Laws of indices for all rational exponents.                                                                   | $a^m \times a^n = a^{m+n}$ , $a^m \div a^n = a^{m-n}$ , $(a^m)^n = a^{mn}$<br>The equivalence of $a^{\frac{m}{n}}$ and $\sqrt[n]{a^m}$ should be known.                                                                                                                                                                                                                                                                      |
| 1.2                      | Use and manipulation of surds.                                                                                | Students should be able to rationalise denominators.                                                                                                                                                                                                                                                                                                                                                                         |
| 1.3                      | Quadratic functions and their graphs.                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.4                      | The discriminant of a quadratic function.                                                                     | Need to know and to use<br>$b^2 - 4ac > 0$ , $b^2 - 4ac = 0$ and $b^2 - 4ac < 0$                                                                                                                                                                                                                                                                                                                                             |
| 1.5                      | Completing the square. Solution of quadratic equations.                                                       | Solution of quadratic equations by factorisation, use of the formula, use of a calculator and completing the square.<br>$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$                                                                                                                                                                                                                 |
| 1.6                      | Solve simultaneous equations; analytical solution by substitution.                                            |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.7                      | Interpret linear and quadratic inequalities graphically.                                                      | For example,<br>$ax + b > cx + d$ , $px^2 + qx + r \geq 0$ , $px^2 + qx + r < ax + b$ .<br>Interpreting the third inequality as the range of $x$ for which the curve<br>$y = px^2 + qx + r$ is below the line with equation $y = ax + b$ .<br>Including inequalities with brackets and fractions. These would be reducible to linear or quadratic inequalities,<br>e.g. $\frac{a}{x} < b$ becomes $ax < bx^2$ , $x \neq 0$ . |
| 1.8                      | Represent linear and quadratic inequalities graphically.                                                      | Represent linear and quadratic inequalities such as $y > x + r$ and $y > ax^2 + bx + c$ graphically.<br>Shading and use of dotted and solid line convention is required.                                                                                                                                                                                                                                                     |
| 1.9                      | Solutions of linear and quadratic inequalities.                                                               | For example,<br>solving $ax + b > cx + d$ , $px^2 + qx + r \geq 0$ ,<br>$px^2 + qx + r < ax + b$ .                                                                                                                                                                                                                                                                                                                           |
| 1.10                     | Algebraic manipulation of polynomials, including expanding brackets and collecting like terms, factorisation. | Students should be able to use brackets. Factorisation of polynomials of degree $n$ , $n \leq 3$ , e.g. $x^3 + 4x^2 + 3x$ . The notation $f(x)$ may be used.                                                                                                                                                                                                                                                                 |

| What students need to learn:                     |                                                                                                                                                                                                         | Guidance                                                                                                                                                                                                                                                                                                                                                |
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| <b>1. Algebra and functions</b> <i>continued</i> |                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                         |
| 1.11                                             | Graphs of functions; sketching curves defined by simple equations. Geometrical interpretation of algebraic solution of equations. Use of intersection points of graphs of functions to solve equations. | <p>Functions to include simple cubic functions and the reciprocal functions</p> $y = \frac{k}{x} \text{ and } y = \frac{k}{x^2} \text{ with } x \neq 0.$ <p>Knowledge of the term asymptote is expected.</p> <p>Also, trigonometric graphs.</p>                                                                                                         |
| 1.12                                             | Knowledge of the effect of simple transformations on the graph of $y = f(x)$ as represented by $y = af(x)$ , $y = f(x) + a$ , $y = f(x + a)$ , $y = f(ax)$ .                                            | <p>Students should be able to apply one of these transformations to any of the above functions (quadratics, cubics, reciprocals, sine, cosine, and tangent) and sketch the resulting graphs.</p> <p>Given the graph of any function <math>y = f(x)</math>, students should be able to sketch the graph resulting from one of these transformations.</p> |